SoniX – Battery Calculations

Current Consumption and Estimated Battery Runtime

1. Current Consumption by Each Component

No.	Component	Current (mA)
1	MPU6050	1
2	MLX90614 (IR non-contact temperature sensor)	2
3	VEML6075 UV sensor	0.48
4	MQ-7 gas sensor	150
5	Buzzer	10
6	ESP32-C3 Mini (Wi-Fi ON)	150

2. Total Current Consumption

Total current = 1 + 2 + 0.48 + 150 + 10 + 150 = 313.48 mA

3. Battery Specification

Battery capacity: 2500 mAh

Nominal battery voltage: 3.7 V (Li-ion)

Maximum Charged Voltage:4.2V(Li-ion)

4. Estimated Battery Runtime

Ideal runtime = Battery capacity (mAh) ÷ Total current (mA)

Ideal runtime = 2500 ÷ 313.48 = 7.97 hours

Note: This is an ideal estimate based on the current values supplied in the handwritten calculation. Actual runtime may be lower because of ESP32 Wi-Fi peaks, sensor warm-up/current variation, voltage-conversion losses, battery condition, and alert activity. The MQ-7 module may draw substantial heater current; verify the exact module's current and supply requirements before finalizing the battery design.

5. Calculation Summary

Total listed current	313.48 mA
Battery capacity	2500 mAh
Ideal calculated runtime	7.97 hours
Practical runtime	To be confirmed by prototype testing